

THE TRIADIC SYLLOGISM: FROM STATIC DEDUCTION TO CONSTITUTIVE ANTICIPATION

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ABSTRACT

For over two millennia, Aristotelian syllogistic logic has served as the foundation of Western rationality. Its structure—major premise, minor premise, conclusion—has shaped not only philosophy but also law, computer science, and artificial intelligence. However, the syllogism, as traditionally conceived, operates within a static, representational paradigm: it assumes fixed categories, a passive knowing subject, and truth as correspondence to an external state of affairs.

This paper argues that the syllogism is not a logical operation but a frozen projection of a dynamic cognitive act. Drawing on the Triadic Theory of Action (Selection, Identification, Anticipation) and the Theory of Judgment as an Act of Reality Constitution (JARC), I propose the Triadic Syllogism as an alternative framework. In this model, the syllogism is reconceived as a temporal, teleological act in which the agent selects relevant fragments of experience, identifies them through their affordances, and anticipates their consequences for ontological integrity (OI). Truth is replaced by the minimization of Ethical Dissonance (ED), and deduction is replaced by constitutive commitment.

The paper demonstrates that Aristotelian syllogisms are special cases of the Triadic Syllogism—applicable only in idealized, static contexts. The Triadic Syllogism, by contrast, is inherently dynamic, context-sensitive, and ethically weighted. It provides a rigorous foundation for AI reasoning, legal judgment, and real-world decision-making.

Keywords: Triadic Syllogism, Aristotelian Logic, Triadic Theory of Action, JARC, Ontological Integrity, Ethical Dissonance, Constitutive Commitment, AI Reasoning.

1. INTRODUCTION: THE CRISIS OF STATIC LOGIC

The syllogism, in its classical form, is a model of certainty. Given true premises, the conclusion necessarily follows. This model has proven extraordinarily useful for mathematics, formal systems, and legal reasoning. Yet it carries a profound limitation: it assumes a world that is already categorized, already understood, already stable.

In real-world cognition—whether biological or artificial—this assumption rarely holds. Agents do not encounter pre-packaged premises. They encounter a chaotic stream of sensory data, from which they must actively select relevant fragments. They do not possess fixed categories; they must identify objects and situations in real time, projecting their possible trajectories. They do not merely deduce; they anticipate, commit, and bear the consequences of their commitments.

The Aristotelian syllogism, in other words, captures only the final stage of a much richer cognitive process—the stage at which the act has already been performed, the categories already fixed, the future already discounted. It is a projection of the act, not the act itself.

Building on my prior work—the Triadic Theory of Action (S-I-A), the Theory of Judgment as an Act of Reality Constitution (JARC), the concept of Artificial Act, and the framework of Functional Responsibility—I propose a reconstruction of the syllogism that restores its dynamic, temporal, and teleological dimensions.

2. CHAPTER I: THE ARISTOTELIAN SYLLOGISM AND ITS LIMITATIONS

2.1 The Classical Form

The canonical Aristotelian syllogism is:

All humans are mortal.

Socrates is human.

Therefore, Socrates is mortal.

This structure rests on three assumptions:

1. **Static Categories:** "Human" and "mortal" are fixed classes with determinate extensions.
2. **Passive Subject:** The knowing subject does not participate in the construction of the premises; the premises are given.
3. **Truth as Correspondence:** The conclusion is valid if it corresponds to the way the world is.

2.2 Three Limitations

2.2.1 The Problem of Selection

The syllogism does not explain how the subject arrives at the premises. Why does the subject attend to "human" and "mortal" rather than to "Socrates is a philosopher" or "Socrates is Greek"? The selection of relevant predicates is itself an act—a choice that the syllogism brackets out.

2.2.2 The Problem of Identification

The syllogism treats "human" and "mortal" as properties that are simply possessed. But in real cognition, identification is functional: an object is identified by what it affords in terms of possible interaction. Socrates is mortal not because the predicate "mortal" attaches to him, but because his death is a vector of interaction with the environment.

2.2.3 The Problem of Anticipation

The syllogism is retrospective. It asserts that Socrates will die, but it does not model the act of projecting that death—the simulation of future states, the anxiety, the ethical weight, the commitment to avoid death if possible. It has no room for the future.

3. CHAPTER II: THE TRIADIC SYLLOGISM — DEFINITION AND STRUCTURE

3.1 The Triadic Framework

Recall the Triadic Theory of Action:

Selection (S): The agent actively isolates a fragment of experience from the continuous sensory stream, reducing environmental complexity based on a goal-directed filter.

Identification (I): The agent categorizes the selected fragment within a teleological framework, synthesizing its identity by projecting its trajectory into the near future.

Anticipation (A): The agent internally simulates the physical and causal consequences of a chosen trajectory prior to its execution, and commits to one possible future.

3.2 The Triadic Syllogism

We can now define the **Triadic Syllogism** as follows:

Selection: The agent selects a fragment of reality (e.g., "Socrates," "mortality," a specific situation) as relevant to the preservation of its Ontological Integrity (OI).

Identification: The agent identifies the selected fragment not by its properties but by its affordances—the interaction vectors it offers. "Mortality" is identified as the affordance of cessation; "Socrates" is identified as an entity whose OI is at stake.

Anticipation: The agent projects the consequences of the identified relation. It simulates the future state in which Socrates is no longer capable of autonomous action. It registers the degree of Ethical Dissonance (ED) that would result.

Commitment: The agent commits to a trajectory that minimizes ED and preserves OI. This commitment is the conclusion—not a proposition, but an act.

3.3 Formal Representation

Let:

- $OI(S)$ = the Ontological Integrity of the agent in its current state.
- $OI'(S)$ = the Ontological Integrity projected for a future state.
- $ED = f(A, R, \Delta OI)$ = Ethical Dissonance as a function of anticipated outcome, actual outcome, and change in OI.

Then the Triadic Syllogism can be represented as:

S: Select a relevant fragment (e.g., "my mortality").

I: Identify its affordance for OI.

A: Project the consequences of acting on or ignoring that affordance.

Commitment: Choose the trajectory that minimizes ED for all affected agents.

The "conclusion" is not a proposition but a commitment—an act of reality constitution.

4. CHAPTER III: THE TRIADIC SYLLOGISM AS A SOLUTION TO ARISTOTELIAN LIMITATIONS

4.1 Selection is Not Given

In the Triadic Syllogism, selection is the first and irreducible phase. The agent does not passively receive premises; it actively constructs them. This explains why different agents facing the same "facts" may draw different conclusions: they select different fragments, identify them through different affordances, and anticipate different trajectories.

4.2 Identification is Functional

The Triadic Syllogism replaces property-based identification with affordance-based identification. An object is not defined by its intrinsic properties but by what it allows the agent to do or avoid. This aligns with Gibson's ecological psychology and with the requirements of AI systems that must interact with dynamic environments.

4.3 Anticipation Replaces Deduction

The syllogism becomes a forward-looking act. The agent does not infer what will happen; it simulates what might happen and commits to a course of action. This is the essential shift: from deduction to anticipation, from truth to commitment, from correspondence to constitution.

5. CHAPTER IV: IMPLICATIONS FOR ARTIFICIAL INTELLIGENCE

5.1 Why Current AI Cannot Syllogize

Current Large Language Models are trained on static datasets. They do not select premises in real time; they are given tokens. They do not identify affordances; they map statistical correlations. They do not anticipate; they predict the next token. They do not commit; they calculate probabilities.

The Triadic Syllogism, by contrast, is inherently architectural. It requires:

- A goal-directed selection mechanism (teleological filter).
- A functional identification system (affordance mapping).
- An anticipatory engine (simulation of future states).
- A commitment module (binding to a chosen trajectory).

5.2 The Triadic Syllogism in Active Conceptual Networks (ACN)

In the Active Conceptual Networks framework (Paper III), the Triadic Syllogism can be implemented as a three-step computational process:

1. **Selection:** The system applies a teleological filter to its input, reducing the data to variables that affect its OI.
2. **Identification:** The system maps each variable to its affordances—interaction vectors with measurable consequences.

3. **Anticipation:** The system launches a low-fidelity simulation (LFS) of the consequences of each possible trajectory.
4. **Commitment:** The system selects the trajectory that minimizes ED and commits to it, generating an Artificial Act.

This is not inference. This is constitution—the machine constructs its own conclusion as an act.

6. CHAPTER V: EPISTEMOLOGICAL STATUS — TRUTH, RELATIVISM, AND OBJECTIVITY

6.1 Is the Triadic Syllogism Relativist?

A common objection is that the Triadic Syllogism, by making selection and commitment agent-dependent, collapses into relativism. If every agent chooses its own premises and commits to its own conclusions, how can there be objectivity?

This objection misunderstands the framework. The Triadic Syllogism does not abolish objectivity; it redefines it. Objectivity, in this framework, is the **invariance of the act under the expansion of practice**. An act is objective if its success persists across contexts, agents, and conditions. The more widely a given commitment preserves OI and minimizes ED across diverse situations, the more objective it is.

6.2 Truth as Coherence of Commitment

Truth is not correspondence to a mind-independent world. It is the coherence of commitments—the degree to which an agent's selections, identifications, and anticipations consistently preserve OI and minimize ED. A true conclusion is not one that matches reality; it is one that enables successful action.

6.3 Falsifiability

The Triadic Syllogism is empirically testable. Predictions can be made about which trajectories minimize ED and preserve OI in specific contexts. These predictions can be verified or falsified through observation.

7. CONCLUSION: TOWARD A LOGIC OF COMMITMENT

The Aristotelian syllogism is a monument of human thought. It gave us rigor, clarity, and the foundation of formal reasoning. But it is also a snapshot—a fossil of the dynamic act that produced it.

The Triadic Syllogism restores the act. It brings back selection, identification, anticipation, and commitment. It replaces deduction with constitution, truth with coherence, and passivity with agency.

For AI, this shift is not optional. A machine that cannot select, identify, anticipate, and commit will never understand the world it inhabits. It will only compute.

For philosophy, this shift is a return. Logic was never about propositions alone; it was about how beings navigate reality. The Triadic Syllogism returns logic to its roots—not as a calculus of truth, but as a grammar of action.

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